MIL-C-9940A
20 JUNE 1988
SUPERSEDING
MIL-C-9940 (USAF)
23 January 1962

MILITARY SPECIFICATION

CONTROL, AUXILIARY HOIST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers auxiliary crane and hoist control units for direct and remote control, precision raising and lowering equipment.

1.2 Classsification. The control units covered by this specification shall have the following load capacities:

Size 1 - 1 Ton
Size 2 - 5 Ton
Size 3 - 10 Ton
Size 4 - 20 Ton

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood Cleated - Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	Preservation, Methods of.
DOD-D-1000	Drawings, Engineering and Associated Lists.
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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STANDARDS

FEDERAL

FED-STD-H28	Screw - Thread Standards for Federal Services.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-889	Dissimilar Metals.
MS28889	Valve, Air, High Pressure Charging.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM D 3951	Packaging, Commercial

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1961 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.5 and 6.3).

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3.2 Components. The control unit shall consist essentially of a reservoir, hydraulic pump, actuating cylinder, accumulator, gauges, upper and lower connection eyes, remote control apparatus, relief valve, and storage case.

3.2.1 Reservoir. The reservoir shall have a minimum hydraulic fluid capacity of one half gallon.

3.2.2 Hydraulic pump. The hydraulic pump shall be manually operated, a high pressure, single acting type, suitable for the rated and test capacity of the unit.

3.2.3 Actuating cylinder. The actuating cylinder with extendable piston, shall be of high grade steel and free from welds and seams. Sliding friction surfaces shall be hard surface plated and polished as required to assure long life and smooth leakage-free operation.

3.2.3.1 Vent valve. A breather port incorporating a dust filter shall be provided in the top of the actuator to vent the portion of the actuating cylinder above the piston to atmosphere.

3.2.4 Accumulator. A hydraulic accumulator shall be provided which shall function as a return air "spring" for the actuator piston. Means shall be provided to prevent entry of the hydraulic fluid into the accumulator, regardless of the position of the unit.

3.2.4.1 Accumulator valve. The accumulator shall be provided with an air charging and relief valve and hose connector conforming to MS28889.

3.2.5 Gauges. Two gauges shall be provided on each control unit. The gauges shall be bourdon tube type, calibrated to read in pounds. They shall be able to withstand 25 percent overload of control unit capacity without damage to components or accuracy. The gauges shall be accurate to plus or minus one percent of full scale.

3.2.5.1 Accumulator gauge. The accumulator gauge shall indicate the force exerted on the actuator piston by the accumulator pressure. The gauge shall be calibrated to read in 100 pound increments.

3.2.5.2 Actuator gauge. The actuator gauge shall indicate the force on the piston of the actuator due to the suspended load, and shall be calibrated in the following increments:

Size 1 - 10 Pound increments
Size 2 - 50 Pound increments
Size 3 - 100 Pound increments
Size 4 - 200 Pound increments

3.2.6 Upper and lower connection eyes. The connection eyes shall be of high grade steel, machined to a good commercial finish.

3.2.7 Remote control apparatus. Remote control apparatus shall be provided and shall consist of lanyards with means of attaching to the pump and relief handles on one end, and a storage reel-type handgrip on the other. The length of the lanyard shall be adjustable, and shall be at least 30 feet long when fully unreeled from the handle.

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3.2.8 Relief Valve. A manual relief valve shall be provided to relieve the pressure in the actuator cylinder. The relief valve shall closely control the flow of fluid. The valve shall throttle the flow such that the rate of piston travel shall depend upon and be approximately proportional to the valve setting. Maximum piston travel with the valve full open and the unit fully loaded shall be approximately 10 feet per minute.

3.2.9 Handles. The pump and relief valve shall handles spring loaded to the up and closed positions. The handles shall be suitable for convenient hand operation and shall incorporate provisions for the attachment of the remote control lanyards.

3.2.10 Storage case. A storage case shall be provided for containing and transporting the complete control unit. The storage case shall be constructed of metal or hardwood and shall be outfitted with corrosion resistant hardware.

3.2.10.1 Castered stands. Castered stands suitable for accomodating the storage case with units, shall be provided with control units of sizes 3 and 4.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. They can be replaced by standard parts (MS and AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the drawings.

3.3.2 Standard parts. With the exception of 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Material. Materials used in construction shall be of such quality as to insure conformance of the control units to requirements specified herein.

3.5 Design. The control units shall be hydraulically operated and entirely self contained, requiring no external connections to hydraulic, electric, or air supplies during operation. The control units shall be suitable or use in any plane, horizontal through vertical. The maximum hydraulic working pressure of the unit shall not exceed 1800 psi.

3.5.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in initimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5.2 Corrosion resistance. Materials selected shall be corrosion resistant or shall be suitably protected by plating, painting, or other surface treatment to resist corrosion.

3.5.3 Threads. All threaded parts shall comply with FED-STD-H28.

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3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern manufacturer's part numbers.

3.7 Construction. The control unit shall be of safe construction, and shall have a minimum structural safety factor of 5 in the extended position. Pressure vessel minimum safety factor is 3. The construction shall be free of any characteristics or defects that may render the unit unsuitable or unsafe for the intended use.

3.7.1 Configuration. The configuration of the unit shall be similar to Figure 1, although Figure 1 is for illustrative purposes only and is not intended to be restrictive (see Figure 1).

3.8 Performance. The control units shall be suitable for operation by one man and have the following performance characteristics.

3.8.1 Positioning. The control units shall be capable of raising and lowering its specified capacity through a 12 inch range.

3.8.2 Piston travel. The actuator piston shall have a travel between .002 and .005 inches per full pump stroke. The travel shall be easily controllable to .001 inches with partial strokes of the pump.

3.9 Seals. All static and moving seals shall be of "O Ring" type and suitable for use with hydraulic oil conforming to MIL-H-5606.

3.10 Safety devices. Each control unit shall be provided with safety devices. Parts which are hazardous to the operator, such as high pressure lines and fittings, shall be well guarded.

3.11 Physical properties. The control units shall conform to the following:

Size	Maximum length (closed pos)	Capacity (tons)	Maximum weight
1	28 inches	1	125 lb
2	28 inches	5	125 lb
3	35 inches	10	320 lb
4	44 inches	20	720 lb

3.12 Marking. The pump and relief valve handles shall be suitably marked indicating control function. The gauges shall be marked appropriately, such as "accumulator preload" and "load weight."

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Painting. Unless otherwise specified in contract or order, (see 6.2) surfaces of the control unit not otherwise finished or plated, and for which a painted surface will be suitable, shall be finished and painted in accordance with good commercial standards. Government specifications for production processes shall be used whenever they are applicable.

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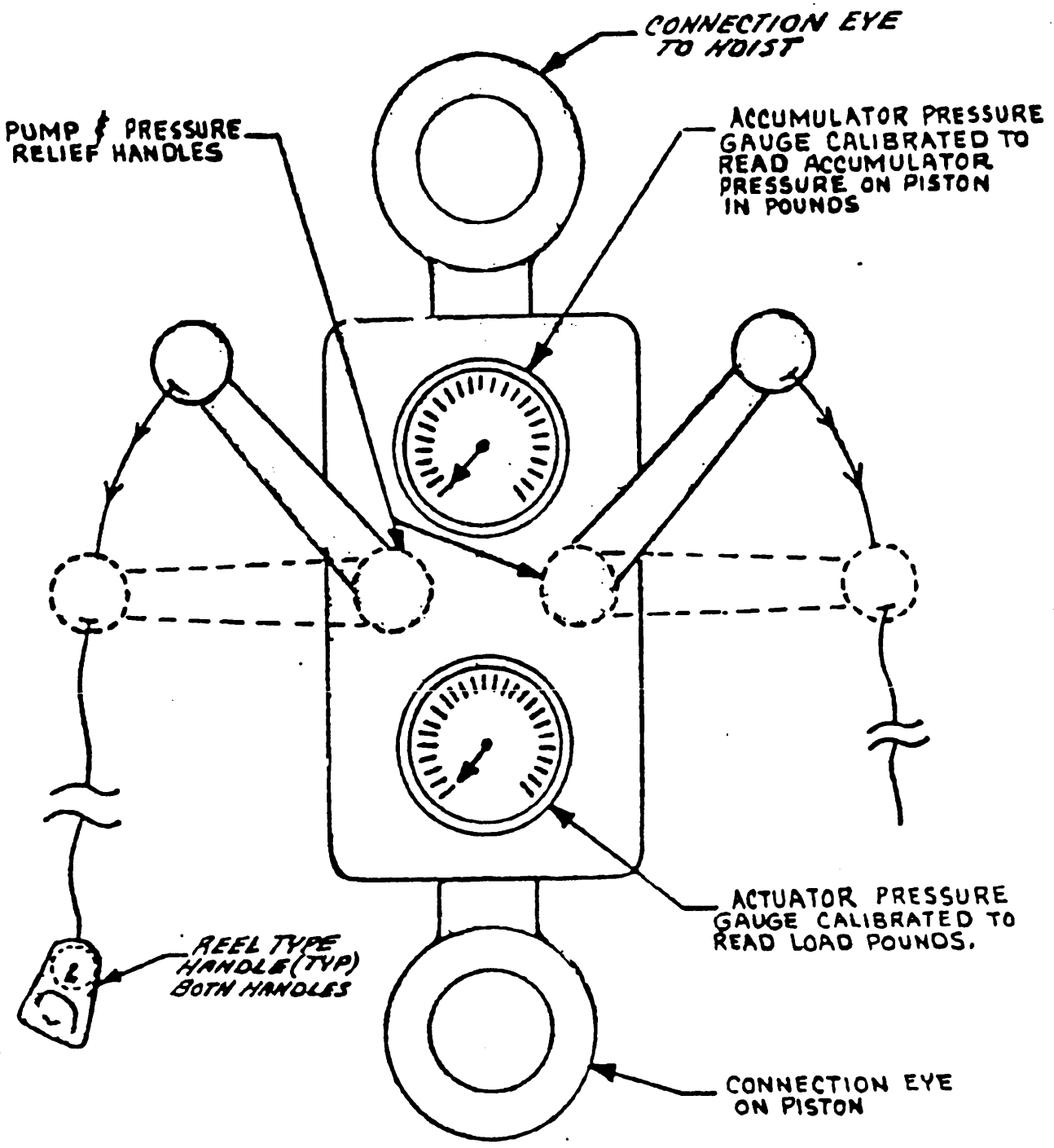


FIGURE 1.

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3.15 Workmanship. All workmanship shall be first class to assure procurement of good quality equipment of a neat general appearance. Standards of workmanship shall be such that equipment specified herein shall comply with the requirements of section 3. All components, subassemblies, parts, and the final assembly shall comply with the test in section 4.

3.16 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 First article inspection. First article inspection shall consist of all the tests specified in 4.4, and examination in accordance with 4.3.1. First article inspection shall be performed by the contractor. Failure of the first article sample to pass the first article inspection shall be cause for rejection.

4.3 Acceptance inspection. Acceptance inspection shall consist of the examination specified in 4.3.1 and the tests specified in 4.3.2. An acceptance lot shall consist of all control units from one contractor submitted for acceptance at essentially the same time.

4.3.1 Acceptance examination. Each control unit submitted for acceptance shall be examined in accordance with MIL-STD-105 using the following classification of defects (see 4.3.1.1) and acceptable quality level (AQL).

4.3.1.1 Classification of defects.

Major defects (if the control unit has one or more major defects, it shall be rejected).

101. Control unit not complete (see 3.2).
102. Evidence that the equipment will not operate (see 3.5).
103. Defective fastening devices affecting operation (see 3.4).
104. Safety devices not as specified (see 3.7).
105. Performance not as specified (see 3.8).

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Minor defects (AQL shall be 4.0 percent defective).

201. Painting not as specified (see 3.14).
202. Marking not in accordance with MIL-STD-130 (see 3.13).
203. Packaging and packing not as specified (see section 5).

4.3.2 Acceptance tests. From the samples which have passed the acceptance examination specified in 4.3.1, samples for acceptance tests shall be selected in accordance with inspection level II of MIL-STD-105. Each sample shall be selected and subjected to the tests a and b below. Acceptance and rejection shall be based on an AQL of 4.0 percent defective.

a. Operational tests (see 4.4.3).

b. Preparation for delivery test (see 4.4.4).

4.4 Test methods.

4.4.1 Load test. The control unit shall support the test load corresponding to twice its rated capacity for one hour. The unit shall be in a vertical position. The load shall be applied either by a weight or a testing machine. CAUTION: All pressure lines to the gauges shall be disconnected and plugged for this test.

4.4.2 Load sustaining test. Each control unit shall sustain for 72 hours the test load corresponding to its capacity. The unit shall show no evidence of leakage or impending failure during or after the test.

4.4.3 Operational test. The unit shall be cycled 25 times through its full range of travel with a capacity load attached. During the test, the units conformance with the operational requirements of paragraphs 3.2.8 and 3.8 shall be determined. At least 5 of the cycles shall be accomplished with the unit in a vertical or near vertical position. During the test the unit shall demonstrate its ability to give the operator complete and accurate control over the position of the load at all times.

4.4.4 Preparation of delivery test. The preservation, packaging, packing, and marking of the control unit shall be examined and tested to determine conformance with the requirements of section 5 of this specification, and as required by specification referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with MIL-STD-105, with inspection level L-5, and an AQL of 4.0 percent defective.

4.5 First article testing. One(1) control (see 3.1 and 6.2) shall be tested for design approval at the contractor's facility under the supervision of the procuring activity.

4.5.1 First article test report. The contractor shall prepare a first article test report in accordance with MIL-STD-831 and furnish 3 copies of the report to the procuring activity.

4.5.2 First article tests. The first article tests shall consist of all tests described under 4.4, conducted in the same manner.

5. PACKAGING

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5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. the control unit shall be thoroughly cleaned of any residue that may cause corrosion on any part of the item. Cleaning fluids used in the cleaning process shall be removed and the item shall be thoroughly dried with dry compressed air or a clean lint-free cloth.

5.1.1.2 Packaging. The control unit shall be packaged in its carrying and storage case with sufficient cushioning and wrapping materials to protect all glass parts, control levers, etc. The item shall be packaged in accordance with method II, MIL-P-116 using the case for the immediate container.

5.1.2 Commercial.

5.1.2.1 Cleaning. The control unit shall be cleaned and dried in accordance with 5.1.1.

5.1.2.2 Commercial packaging. The control unit shall be packaged within its carrying and storage case in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.2.1 Level A. The packaged controls shall be packed one each in containers conforming to PPP-B-621 or PPP-B-601 overseas type. Closure and strapping shall be in accordance with applicable container specification or appendix thereto.

5.2.2 Level B. The packaged controls shall be packed one each in containers conforming to PPP-B-621 or PPP-B-601 domestic type. Closure and strapping shall be in accordance with applicable container specification or appendix thereto.

5.2.3 Commercial. The packaged controls shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.2) each unit pack and exterior container shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The control units covered by this specification are intended for use as a link between a crane or hoist line and a load. The control unit shall be used for direct or remote control, precision raising and lowering of equipment.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Painting and finishing of control unit and storage case (see 3.14).

d. Level of packing and packaging required (see section 5).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word listing).

Control, precision
Auxiliary load indicator
Hoist
Positioner weight or strain

* 6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AL
Air Force - 99

Review activities:
DLA - CS
Navy - YD

Preparing activity:
Air Force - 82

(Project 4940-F488)

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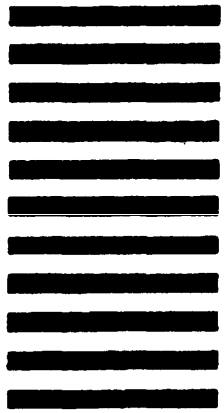


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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL C 9940 A	CONTROL, AUXILIARY HOIST

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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